Positive Lower Density of Simultaneous Primitive Roots

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1 Statement

For a prime p not dividing an integer a, write $\mathrm{ord}_p(a)$ for the multiplicative order of a modulo p. For a finite set $A \subset \mathbb{N}_{\geq 2}$, put

$$N_A(x) = \#\{p \text{ prime} : x < p \leq 2x, \quad p \nmid \textstyle\prod_{a\in A} a, \quad \mathrm{ord}_p(a) = p-1 \text{ for every } a \in A\}.$$

Theorem 1.1. *Assume the analytic estimates in Section 2. Let $A \subset \mathbb{N}_{\geq 2}$ be a nonempty finite set such that*

$$\prod_{a\in B} a \text{ is not a square whenever } B \subseteq A \text{ has odd cardinality.} \tag{1.1}$$

Then there are constants $c_A > 0$ and $x_A \geq 2$ such that

$$N_A(x) \geq c_A \frac{x}{\log x} \qquad (x \geq x_A). \tag{1.2}$$

Conversely, if (1.1) fails, no odd prime has every element of A as a primitive root.

In particular, (1.2) holds for $A = \{2, 3\}$ and for $A = \{3, 6\}$, and implies positive lower density relative to the primes.

All letters denoting primes range only over primes. Constants may depend on parameters that have already been fixed. We use $L = \log x$ and $\mathrm{e}(t) = \exp(2\pi i t)$.

2 Analytic inputs

The following analytic estimates are taken as inputs.

2.1 Marked correlation estimates

Fix $1/2 < \alpha < \beta < 1$, and define

$$f_x(n) = \sum_{x^\alpha < Q \leq x^\beta} \mathbf{1}_{Q|n}, \qquad C_x = \sum_{x^\alpha < Q \leq x^\beta} \frac{1}{Q}. \tag{2.1}$$

On every range $n \asymp x^{1+o(1)}$, the functions f_x and C_x are bounded uniformly in x.

A marking array consists of K disjoint prime bands

$$\mathcal{P}_i = \{s : \exp(L^{a_i}) \leq s \leq \exp(2L^{a_i})\}, \qquad 0.1 < a_1 < \cdots < a_K < 0.2,$$

where K and the exponents are fixed. Write

$$V_i = \sum_{s\in\mathcal{P}_i} \frac{1}{s}, \qquad \omega_i(n) = \sum_{s\in\mathcal{P}_i} \mathbf{1}_{s|n}, \qquad \mathcal{W}(n) = \prod_{i=1}^{K} 2^{1-\omega_i(n)} \frac{\omega_i(n)}{V_i}.$$

The value is zero if some $\omega_i(n)$ is zero. Each such weight is bounded for fixed K. Let M_x be the divisor model in which the band-prime divisibility indicators are independent Bernoulli variables with probabilities $1/s$, and put $m_{\mathcal{W}} = M_x(\mathcal{W})$. We use

$$0 < c_K \le m_{\mathcal{W}} \le C_K, \qquad M_x(\mathcal{W}^2) \le C_K, \tag{2.2}$$

with bounds independent of the particular fixed band exponents.

Here are the correlation inputs and the features of them needed in this proof.

(M1) **One-sided estimate.** Put $W = \exp(L^{0.24})$ and $V(W) = \prod_{s\le W}(1-1/s)$. For $H_m, H_n \ge x^{\varepsilon}$, $H_m H_n \asymp x$, let

$$\alpha_m = m^{it}\left(\mathbf{1}_{m \text{ prime}} - \frac{\mathbf{1}_{P^-(m)>W}}{V(W)\log m}\right), \qquad |t| \le L^B,$$

on any subinterval of $[H_m, 2H_m]$. Let β_n be supported on W-rough integers in $[H_n, 2H_n]$, with $|\beta_n| \le L^B$. If $|F| \le 1$ and $F(sn) = F(n)$ for every band prime s, then, for every prescribed A_0, sufficiently large fixed K gives

$$\sum_{m,n} \alpha_m \beta_n F(mn-1)\mathcal{W}(mn-1) \ll xL^{-A_0}. \tag{2.3}$$

The required K is independent of the fixed band exponents. We also use the coefficient form of this estimate: it holds for W-rough-supported coefficients $\widetilde{\alpha}_m$ with a fixed logarithmic bound whenever, for every fixed A_1, B_1,

$$H_m^{-1}\left|\sum_m \widetilde{\alpha}_m \chi(m) m^{iu}\right| \ll L^{-A_1}, \qquad |u| \le 2H_m H_n L^{B_1}, \tag{2.4}$$

uniformly for all characters of any prescribed logarithmically bounded modulus and all subintervals of the dyad. The prime-minus-rough coefficients α_m above satisfy (2.4), including for principal and imprimitive characters. The logarithmic precision and modulus and time bounds are fixed before K is chosen.

(M2) **Two-sided transfer and endpoint estimate.** Take pairwise disjoint active prime groups $\mathcal{P}_g$, with between fixed positive multiples of $\log L$ groups of each of two kinds, the small groups lying in $[\exp(L^{0.27}), \exp(L^{0.36})]$ and the big groups in $[\exp(L^{0.39}), \exp(L^{0.46})]$. Their reciprocal masses $V_g = \sum_{s\in\mathcal{P}_g} 1/s$ are bounded above and below by fixed positive constants. Big groups are prime intervals. These groups are disjoint from the array bands. Write $\omega_g(n) = \sum_{s\in\mathcal{P}_g} \mathbf{1}_{s|n}$ and $\omega_P(n) = \sum_g \omega_g(n)$. A represented list contains t_g ordered distinct divisors of n from each group g. For a bounded function $\boldsymbol{b}$ of these lists, put

$$W_{\boldsymbol{t}}^{\boldsymbol{b}}(n) = 2^{-\omega_P(n)+\sum_g t_g} \prod_g V_g^{-t_g} \sum_{\text{represented lists at } n} \boldsymbol{b}(\boldsymbol{s}).$$

The weight is zero when no such list exists. Write W_1 when every $t_g = 1$ and $\boldsymbol{b} = 1$.

The marked two-sided correlation method applies to the centered first endpoint $F_x(n) = \mathcal{W}(n)(f_x(n) - C_x)$, in the sum

$$\sum_{n\ge 1} \frac{\Psi(n/x)}{n} F_x(n) G(n+1) W_1(n) W_1(n+1). \tag{2.5}$$

Here Ψ is fixed, smooth and compactly supported, $G(n) = G(n_*)$, and $|G(n)| \le L^B \tau(n_*)^B$, where n_* is obtained by removing all active prime factors with their multiplicities, and τ is the divisor-counting function. Its graph transfer allows bounded position-dependent vectors at both endpoints, independent of the chosen marks. Multiplying either endpoint by a function of modulus at most one preserves its norm and absolute-error bounds. The ideal kernel is unchanged. Shared prime labels are divided out of both endpoints; the resulting joint endpoint factor must be invariant under this common division. The transfer is bilinear in the endpoint vectors.

The transfer expresses the unit-shift correlation as a residual term with arbitrary logarithmic saving and a fixed-power-of-L combination of comparison correlations. In a comparison correlation the first endpoint has the form

$$\mathcal{U}(n) = \psi(n/X) n^{i\sigma} (f_x(n) - C_x) \mathcal{W}(n) W_{\boldsymbol{t}}^{\boldsymbol{b}}(n).$$

Its scale has the form

$$X = xH', \qquad H' = U_1 U_2, \qquad U_i \ge \exp(L^{0.39} - O(1)), \qquad H' \le \exp(O(L^{0.46} \log L)).$$

The two scales U_i come from nonempty free products of big-group labels. Here the t_g are bounded fixed positive integers, with $t_g = 1$ on every small group. The list test $\boldsymbol{b}$ depends only on big-group marks and has modulus at most one, and σ has a fixed logarithmic bound. The active primes are outside the range in (2.1). We use the uniform estimate

$$\int_{-T}^{T} \left| X^{-1} \sum_n \mathcal{U}(n) n^{it} \mathrm{e}(hn/k) \right|^2 \mathrm{d}t \ll_{A_0,B,C} L^{-A_0}, \quad k \le L^B, \quad T \le xL^C, \tag{2.6}$$

for arbitrary fixed A_0, B, C, including frequency zero. Fixed logarithmic twists and smooth cutoffs are allowed. Together with the second endpoint's global square norm and the comparison multiplier bound, this estimate gives arbitrary logarithmic saving for each comparison. The absolute Schur bounds and all finite decomposition costs are fixed logarithmic powers, chosen before the saving in (2.6). Subtracting the comparison terms from the residual term bounds (2.5) by $O(L^{-A_0})$ for every fixed A_0.

(M3) **Prime extraction.** The one-sided and two-sided estimates apply in the following presieve and extraction procedure. A presieve with depth h, level x^{c_0}, and cutoff

$$z = x^{\gamma}, \qquad \gamma = \frac{c_0}{4h+3}, \tag{2.7}$$

has relative error $O(e^{-h})$ plus its summed progression remainders. Here $0 < c_0 < (1-\beta)/2$ is fixed. For the composite terms that remain, prime factors are replaced by the rough coefficients in (2.3). A nonnegative partition into successful marking configurations, followed by allocation of marks, converts these terms into the two-sided correlations in (M2). Discarded configurations have total absolute mass $o(x/L)$ before any signed expansion. All subsequent expansion costs are fixed powers of L. The resulting prime estimate is normalized by $m_{\mathcal{W}}$; its surviving presieve-error constant is independent of K, h and the particular fixed array. Progression remainders may have constants depending on these fixed parameters.

2.2 Zero-free input and prime ideals

We also use a fixed $\delta \in (0, 1/2)$ such that, for every cyclotomic number field F containing μ_{12} and every finite-order Hecke character χ of F,

$$L_F(s, \chi) \ne 0 \qquad (\Re s > 1 - \delta), \tag{2.8}$$

with the principal pole at $s = 1$ permitted. The same δ works for all these fields and characters. One may take the input value $\delta = 10^{-6}$.

We use the usual smooth explicit formula for Dedekind zeta functions, with its dependence on degree and discriminant retained. If K has degree n_K, absolute discriminant D_K, and no zeta zero in $\Re s > 1 - \delta$, then, for fixed nonnegative $g \in C_c^\infty((0,\infty))$,

$$\sum_{\mathfrak{p}} \sum_{j\ge 1} (\log N\mathfrak{p}) g\left(\frac{(N\mathfrak{p})^j}{x}\right) \ll_g x + x^{1-\delta}(\log D_K + n_K). \tag{2.9}$$

The constant is independent of K and x. We also use the prime number theorem in fixed reduced residue classes, Mertens' prime reciprocal asymptotic, quadratic reciprocity, and elementary properties of cyclotomic and Kummer extensions.

3 A fixed-divisor statistic

For a fixed integer $D \ge 1$, write $\Delta_D(n) = \mathbf{1}_{D|n}$.

Lemma 3.1 (Finite projection)**.** *For every finitely supported complex sequence (a_n) on the positive integers and every $T \ge 0$,*

$$\sum_n \Delta_D(n) a_n n^{it} \mathrm{e}(\theta n) = \frac{1}{D} \sum_{j=0}^{D-1} \sum_n a_n n^{it} \mathrm{e}((\theta + j/D)n), \tag{3.1}$$

$$\int_{-T}^{T} \left| \sum_n \Delta_D(n) a_n n^{it} \mathrm{e}(\theta n) \right|^2 \mathrm{d}t \le \frac{1}{D} \sum_{j=0}^{D-1} \int_{-T}^{T} \left| \sum_n a_n n^{it} \mathrm{e}((\theta + j/D)n) \right|^2 \mathrm{d}t. \tag{3.2}$$

If $(s, D) = 1$, then $\Delta_D(sn) = \Delta_D(n)$.

Proof. The finite geometric sum $D^{-1} \sum_{j=0}^{D-1} \mathrm{e}(jn/D)$ equals 1 when $D \mid n$ and 0 otherwise. This proves (3.1). The inequality $|D^{-1} \sum_j z_j|^2 \le D^{-1} \sum_j |z_j|^2$, followed by integration, proves (3.2). Finally, coprimality gives $D \mid sn$ if and only if $D \mid n$. □

Proposition 3.2. *Under the marked analytic inputs of Section 2, for fixed D, α, β and every nonnegative $\Phi \in C_c^\infty((0,\infty))$,*

$$\sum_p \Phi((p-1)/x) \Delta_D(p-1)(f_x(p-1) - C_x) = o(x/\log x). \tag{3.3}$$

Proof. We keep D fixed throughout. In particular, all marking primes are coprime to D once x is sufficiently large.

The two-sided endpoint. Multiply the first endpoint by Δ_D. Its supremum and square norm do not increase, and it remains independent of the selected marks. If R is a product of shared marking labels, $\Delta_D(Rn) = \Delta_D(n)$, so division by R preserves the factor. The ideal operator and its coefficients have not changed. All absolute collision and excluded-label errors are bounded by their old estimates.

For a comparison first endpoint $\mathcal{U}$, apply Lemma 3.1 with $a_n = X^{-1}\mathcal{U}(n)$. If $\theta = h/k$, each shifted frequency has denominator at most Dk. For every fixed B, $Dk \le L^{B+1}$ eventually when $k \le L^B$. Thus (2.6), at a larger fixed denominator exponent, bounds every term on the right of (3.2). Choose its logarithmic precision after the fixed comparison costs. The transfer and comparison estimates therefore apply with Δ_D at the first endpoint. This includes shifts that reduce to frequency zero.

The progression main term. For an array weight H, including products of two array weights, expand its band-prime factors into a truncated divisor sum. Truncating at $\lfloor L^{0.01} \rfloor$ additional primes gives divisors $d' \le \exp(O(L^{0.22})) = x^{o(1)}$. The total reciprocal coefficient mass is bounded for each fixed collection of arrays, and the discarded reciprocal mass is smaller than every fixed negative power of L.

In a positive long-prime term, write $n = DQd'v$. All factors of Qd' are coprime to D. If $(r, D) > 1$, the simultaneous conditions $D \mid n$ and $r \mid n+1$ are impossible. If $(r, DQd') = 1$, smooth summation on the single progression gives

$$\sum_{DQd'v \equiv -1 \pmod r} \Phi(DQd'v/x) = \frac{x}{DQd'r} \int_0^\infty \Phi(t)\,\mathrm{d}t + O_\Phi(1). \tag{3.4}$$

If a band prime divides r, its indicator is forced to be zero in the divisor model. The difference from the unconditioned model is at most a fixed constant times $\sum_{s \mid r,\ s \text{ in a band}} 1/s \ll L\exp(-L^{0.1})$. The same calculation without Q handles the constant term.

There are at most x^β choices of Q. Consequently the summed rounding error for $r \le x^{c_0}$ is $O_D(x^{\beta + c_0 + o(1)})$, which saves a fixed power of x. The truncated-model errors and the forced-absence error save arbitrary powers of L. We obtain the two progression main terms

$$\frac{x}{D}\left(\int \Phi\right) M_x(H)\, g_D(r)\, C_x, \qquad \frac{x}{D}\left(\int \Phi\right) M_x(H)\, g_D(r),$$

where $g_D(r) = \mathbf{1}_{(r,D)=1}/r$ for squarefree r.

Presieving. Apply the nonnegative presieve separately to $\Phi(n/x)\Delta_D(n)\mathcal{W}(n)f_x(n)$ and to $\Phi(n/x)\Delta_D(n)\mathcal{W}(n)$. Its local densities are

$$g_D(s) = 0 \ (s \mid D), \qquad g_D(s) = 1/s \ (s \nmid D).$$

For large x, every prime divisor of D is below $z = x^\gamma$, and hence

$$\frac{1}{D}\prod_{s \le z}(1 - g_D(s)) = \frac{1}{\varphi(D)}\prod_{s \le z}(1 - 1/s). \tag{3.5}$$

Subtract C_x times the second formula. Their principal terms cancel. The level condition $z^{4h+2} < x^{c_0}$ holds by (2.7). After dividing by $m_{\mathcal{W}}$, the surviving error is at most

$$C_D \frac{e^{-h}}{\varphi(D)\gamma}\frac{x}{L} + o(x/L), \tag{3.6}$$

where C_D is independent of K, h and the fixed array. Both sieve main terms contain $m_{\mathcal{W}}$, and the relative error is controlled using $g_D(s) \le 1/s$. Thus division by $m_{\mathcal{W}}$ leaves no array-dependent error factor.

Removing composites. Every surviving composite $n+1$ has between 2 and $2/\gamma$ prime factors, counted with multiplicity, for sufficiently large x. Terms with a repeated factor larger than z have total mass $O(x/z)$ times fixed weight bounds, hence $o(x/L)$. For each of the finitely many factorization lengths, use (2.3) to replace the prime slots by rough slots. The required first factor is a bounded normalization of $\Delta_D(n)(f_x(n) - C_x)$. It is invariant under every array prime. A bounded clipping outside the range of interest preserves this invariance.

The marking partition in (M3) discards failure mass before the signed expansion. Multiplication by Δ_D does not increase that absolute mass. Write $v = \prod_i m_i = n+1$. Removing candidate prime parts from the coefficient functions has an absolute error $o(x/L)$. Reallocating those parts among the factors m_i leaves the actual product v unchanged. Thus $\Delta_D(v-1)$ factors out of the allocation sum, and the first endpoint remains $\Delta_D(n)\mathcal{W}(n)(f_x(n) - C_x)$.

The two-sided estimate applies to each retained term. To pass from the correlation measure $1/n$ to the integer sum, put $\Psi(t) = t\Phi(t)$, so that $\Phi(n/x) = x\Psi(n/x)/n$. Choosing the precision after the fixed logarithmic expansion costs gives a total composite contribution $o(x/L)$. Subtracting this from the presieved sum gives

$$\limsup_{x\to\infty} \frac{L}{x} \left| \sum_p \Phi((p-1)/x)\Delta_D(p-1)\frac{\mathcal{W}(p-1)}{m_{\mathcal{W}}}(f_x(p-1) - C_x) \right| \le C_D \frac{e^{-h}}{\varphi(D)\gamma}. \tag{3.7}$$

Removing the array weight. Fix h and a sufficient K, and choose b disjoint arrays. Set

$$Z_b(n) = \left(\frac{1}{b} \sum_{a=1}^{b} \frac{\mathcal{W}_a(n)}{m_a} - 1 \right)^2.$$

Independence of the arrays in the divisor model gives $M_x(Z_b) \le C_K/b$. The progression calculation with $D = 1$, applied to the square and its cross terms, and a fixed-depth upper presieve give

$$\limsup_{x\to\infty} \frac{L}{x} \sum_p \Phi((p-1)/x) Z_b(p-1) \ll C_K/b. \tag{3.8}$$

Here the implied constant is independent of b, K. Average (3.7) over the arrays. Since $\Delta_D \le 1$, Cauchy–Schwarz, (3.8), the boundedness of $f_x - C_x$, and the prime number theorem bound the difference from the unweighted sum by $O(\sqrt{C_K/b})x/L$ in normalized limsup. Therefore

$$\limsup_{x\to\infty} \frac{L}{x} \left| \sum_p \Phi((p-1)/x)\Delta_D(p-1)(f_x(p-1) - C_x) \right| \le C_D \frac{e^{-h}}{\varphi(D)\gamma} + C\sqrt{C_K/b}. \tag{3.9}$$

First let $b \to \infty$ with h, K fixed. Then let $h \to \infty$, choosing a sufficient fixed K each time. Since $\gamma^{-1} = (4h+3)/c_0$, the right side tends to zero. All arrays and parameters are fixed before the x limit. This proves (3.3). □

4 Fixed residue classes

Proposition 4.1. *For fixed $H \ge 1$, $(b, H) = 1$, α, β, and smooth compactly supported $\Phi \ge 0$,*

$$\sum_p \Phi((p-1)/x)\mathbf{1}_{p\equiv b\ (H)}(f_x(p-1) - C_x) = o(x/\log x). \tag{4.1}$$

Proof. We give the changes to Proposition 3.2. Define the periodic two-variable function

$$\mathcal{B}(s,t) = \mathbf{1}_{(t,H)=1}\mathbf{1}_{bs\equiv(b-1)t\ (H)}.$$

Direct simplification gives

$$\mathcal{B}(u, u+1) = \mathbf{1}_{u+1\equiv b\ (H)}, \tag{4.2}$$

$$\mathcal{B}(Rs, Rt) = \mathcal{B}(s,t) \quad ((R,H) = 1), \tag{4.3}$$

$$\mathcal{B}(w, w+d) = \mathbf{1}_{w\equiv(b-1)d\ (H)} \quad ((d,H) = 1). \tag{4.4}$$

For the last identity, the congruence reduces to $w \equiv (b-1)d$, which also gives $(w+d, H) = 1$.

Write $\mathcal{B}$ as a sum of at most H^2 products of residue indicators, one at each endpoint. Insert these into the bilinear graph transfer. The endpoint norms do not increase, and the same ideal kernel applies to every term. Recombine the finite sum before dividing out the

shared labels. Their product R is coprime to H for large x, so (4.3) preserves the joint factor exactly.

In a comparison correlation the shift is the product of the two free label products. Partition each free tuple by its residue modulo H. These bounded tests preserve the endpoint hypotheses. On each part the shift d mod H is fixed, and (4.4) leaves a fixed residue condition on the first endpoint. For such a condition,

$$\mathbf{1}_{n\equiv c\ (H)} = \frac{1}{H}\sum_{j=0}^{H-1} \mathrm{e}(j(n-c)/H).$$

The proof of Lemma 3.1 applies with coefficients $\mathrm{e}(-jc/H)$ of modulus one. The shifted denominators are at most Hk, and (2.6) at a larger fixed logarithmic exponent bounds their full time intervals. The finite splitting cost is absorbed by its arbitrary logarithmic precision. Thus the two-sided estimate holds with $\mathbf{1}_{n+1\equiv b\ (H)}$.

For the one-sided prime replacement, use characters extended by zero on nonunits:

$$\mathbf{1}_{mn\equiv b\ (H)} = \frac{1}{\varphi(H)}\sum_{\lambda \bmod H} \overline{\lambda(b)}\lambda(m)\lambda(n). \tag{4.5}$$

Keep F invariant and place these characters in the two coefficients. If χ has modulus k, then $\lambda\chi$ has modulus $\operatorname{lcm}(H,k) \le Hk$. Since H is fixed, the original coefficient estimate with its modulus exponent increased by one proves (2.4) for $\alpha_m\lambda(m)$, on the same full time interval. The coefficient form of the one-sided estimate therefore applies. Recombining the character sum restores the congruence on the actual product.

For the progression main term, put $n = Qd'u$. The conditions $n+1 \equiv b \pmod H$ and $r \mid n+1$ are impossible if $(r, Hd') > 1$. Otherwise u lies in one class modulo Hr, and its smooth main term is $x\int\Phi/(HQd'r)$. The rounding and forced-absence errors have the same bounds as before. The local sieve density is zero at primes dividing H and $1/s$ elsewhere, so its main factor is $1/\varphi(H)$. The normalized presieve error is $O(e^{-h}/(\varphi(H)\gamma))$.

All discarded masses in the composite extraction are bounded absolutely before the signed expansion. Reallocation preserves the actual product $v = n+1$, hence $v \equiv b \pmod H$. The preceding one-sided and two-sided estimates remove the retained terms. Finally $\mathbf{1}_{p\equiv b\ (H)}Z_b(p-1) \le Z_b(p-1)$, so the same variance bound removes the array weights. Letting x, then the number of arrays, then h tend to infinity proves (4.1). □

Corollary 4.2. *Fix $1/2 < \alpha < \beta < 1$, put $\theta = \log(\beta/\alpha)$, and take $0 \le \Phi \le 1$ smooth with compact support in $(5/4, 7/4)$ and $m_\Phi = \int\Phi > 0$. For every fixed reduced class $b \pmod H$,*

$$\sum_{\substack{p\equiv b\ (H)\\ f_x(p-1)=1}} \Phi((p-1)/x) = \left(\frac{m_\Phi\theta}{\varphi(H)} + o(1)\right)\frac{x}{\log x}.$$

Proof. Use Proposition 4.1, the prime number theorem in the fixed class, and $C_x \to \theta$. Since $2\alpha > 1$, at most one prime larger than x^α divides $p-1 < 2x$ for large x. Thus $f_x(p-1)$ is its own indicator on the support of Φ. □

5 A uniform splitting bound

For a fixed integer $a \ge 2$ and a prime q, let

$$B_{a,q}(x) = \#\{x < p \le 2x : p \nmid aq,\quad q \mid p-1,\quad a^{(p-1)/q} \equiv 1 \pmod p\}.$$

Proposition 5.1. *Under (2.8) and (2.9), there are q_a, C_a such that for all primes $q > q_a$ and all $x \geq 2$,*

$$B_{a,q}(x) \leq C_a \left(\frac{x}{q(q-1)\log x} + \frac{x^{1-\delta}\log(2q)}{\log x} \right). \tag{5.1}$$

The constant C_a does not depend on q or x.

Proof. Let $E_q = \mathbb{Q}(\mu_q)$ and $K_q = \mathbb{Q}(\mu_q, a^{1/q})$. Choose a prime $\ell \mid a$. Once $q \neq \ell$ and $q > v_\ell(a)$, the valuation of a at a prime above ℓ in E_q is not divisible by q. Thus a is not a qth power in E_q, and Kummer theory gives

$$n_q = [K_q : \mathbb{Q}] = q(q-1).$$

The cyclotomic discriminant formula and the discriminant of $X^q - a$ imply

$$D_q \leq q^{q(q-2)}(q^q a^{q-1})^{q-1}, \qquad \log D_q + n_q \ll_a n_q \log(2q).$$

Put $F_q = \mathbb{Q}(\mu_{12q})$ and $\widetilde{K}_q = K_q F_q$. Both $\widetilde{K}_q/F_q$ and $\widetilde{K}_q/K_q$ are abelian. Abelian factorization over F_q and (2.8) show that $\zeta_{\widetilde{K}_q}$ is zero-free in $\Re s > 1 - \delta$, apart from its pole at 1. The quotient $\zeta_{\widetilde{K}_q}/\zeta_{K_q}$ is a product of entire nonprincipal Hecke functions. A zero of ζ_{K_q} in that region would consequently be a zero upstairs. Hence ζ_{K_q} has the same zero-free region.

Choose a fixed $g \geq 0$ in $C_c^\infty((0,\infty))$ with $g \geq 1$ on $[1,2]$. For $p \nmid aq$, the conditions counted by $B_{a,q}$ say that $X^q - a$ splits completely over $\mathbb{F}_p$, or equivalently that p splits completely in K_q. Such a prime contributes at least $n_q \log x$ to the nonnegative left side of (2.9). Division by $n_q \log x$ therefore gives

$$B_{a,q}(x) \ll_g \frac{x}{n_q \log x} + \frac{x^{1-\delta}(\log D_q + n_q)}{n_q \log x},$$

which proves (5.1) uniformly in q. □

6 Removing order obstructions

Lemma 6.1. *Let p be an odd prime and $p \nmid a$. Put $h_a(p) = (p-1)/\operatorname{ord}_p(a)$. For a prime $q \mid p-1$,*

$$q \mid h_a(p) \quad \Longleftrightarrow \quad a^{(p-1)/q} \equiv 1 \pmod{p}.$$

If a is a quadratic nonresidue modulo p, then $h_a(p)$ is odd.

Proof. Writing $d = \operatorname{ord}_p(a)$, the displayed congruence is equivalent to $d \mid (p-1)/q$, which in turn is equivalent to $q \mid h_a(p)$. For $q = 2$ the congruence contradicts Euler's criterion for a quadratic nonresidue. □

Lemma 6.2. *For a fixed $a \geq 2$ and $T \geq 1$, the number of primes $p > x$ with $p \nmid a$ and $\operatorname{ord}_p(a) \leq T$ is*

$$O_a\left(\frac{T^2}{\log x}\right), \qquad x > 1.$$

Proof. Each such prime divides $M = \prod_{1 \leq j \leq \lfloor T \rfloor}(a^j - 1)$. This is a positive integer, and $\log M \leq (\log a)\sum_{j \leq T} j = O_a(T^2)$. If there are r distinct primes in question, their product divides M, so $r \log x \leq \log M$. This also covers $M = 1$. □

Proposition 6.3. *Fix $0 < \eta < \min(\delta, 1/2)$ and a finite set A of bases. Take Y larger than the finitely many thresholds q_a in Proposition 5.1. Consider candidate primes $x < p \le 2x$ satisfying*

$$p - 1 = Qr, \qquad x^{1-\eta} < Q \le x^{1-\eta/2}, \qquad r < 2x^{\eta},$$

every $a \in A$ is a quadratic nonresidue modulo p, and no odd prime $\ell \le Y$ divides $p-1$. Then the number of these candidates at which at least one $a \in A$ fails to be a primitive root is at most

$$\left(\frac{C_A}{Y} + o(1)\right)\frac{x}{\log x}, \tag{6.1}$$

where C_A can be chosen independently of Y.

Proof. Increase the eventual x threshold so that p divides none of the fixed bases. If $h_a(p) > 1$, choose a prime $q \mid h_a(p)$. Lemma 6.1 and the candidate conditions imply $q > Y$.

If $q \ne Q$, then $q \mid r$ and $q < 2x^{\eta}$. Summing (5.1) over this range gives

$$\begin{aligned}\sum_{Y<q<2x^{\eta}} B_{a,q}(x) &\ll_a \frac{x}{\log x}\sum_{q>Y}\frac{1}{q(q-1)} + \frac{x^{1-\delta}}{\log x}\sum_{q<2x^{\eta}}\log(2q)\\ &\ll_a \frac{x}{Y\log x} + x^{1-\delta+\eta}.\end{aligned}$$

The first sum is bounded by its sum over all integers, which telescopes. The second uses at most $2x^{\eta}$ terms, each $O(\log x)$. Since $\eta < \delta$, the second error is $o(x/\log x)$.

If $q = Q$, then $\operatorname{ord}_p(a) \le (p-1)/Q = r < 2x^{\eta}$. Lemma 6.2 bounds this case by $O_a(x^{2\eta}/\log x) = o(x/\log x)$, because $2\eta < 1$. Finally sum the bounds over the fixed set A. No independence between its bases is needed. □

7 Quadratic conditions and completion

Lemma 7.1. *Suppose A satisfies (1.1). There are a fixed modulus H_0 and a reduced class $b_0 \pmod{H_0}$ such that every base in A is a quadratic nonresidue modulo each prime p in that class, and no odd prime dividing H_0 divides $p-1$.*

Proof. Let S consist of $2, 3$ and all prime divisors of $\prod_{a\in A} a$. For $a \in A$ let $v_a \in \mathbb{F}_2^S$ be its vector of prime exponents modulo two. Condition (1.1) says that every linear relation among the v_a has an even number of terms. Consequently $v_a \mapsto 1$ defines a linear functional on their span. Extend it to $\mathbb{F}_2^S$ and write it as $v \mapsto v \cdot z$. Put $s_\ell = (-1)^{z_\ell}$.

Choose a class modulo 24 from the table

(s_2, s_3)	$(-1,+1)$	$(+1,+1)$	$(-1,-1)$	$(+1,-1)$
$p \bmod 24$	11	23	5	17

so that $(2/p) = s_2$, $(3/p) = s_3$ and $p \equiv 2 \pmod 3$. This also determines $\tau = (-1/p)$. For each $\ell \in S$, $\ell \ge 5$, choose $r_\ell \not\equiv 0, 1 \pmod \ell$ with

$$\left(\frac{r_\ell}{\ell}\right) = s_\ell \tau^{(\ell-1)/2}.$$

Such a residue exists: every nonresidue differs from 1, and there is a square residue other than 1 when $\ell \ge 5$. The Chinese remainder theorem gives a reduced class b_0 modulo

$$H_0 = 24\prod_{\substack{\ell\in S\\ \ell\ge 5}} \ell.$$

Quadratic reciprocity gives $(\ell/p) = s_\ell$ for all $\ell \in S$, so $(a/p) = (-1)^{v_a\cdot z} = -1$ for every $a \in A$. The choices modulo the odd primes also give $p \not\equiv 1 \pmod \ell$. □

For fixed $Y \geq 3$, retain the class $b_0 \pmod{H_0}$ and, for every odd prime $\ell \leq Y$ not dividing H_0, retain all residues other than $0, 1$. This is a finite union of reduced classes of a fixed modulus, of relative density

$$\lambda_{A,Y} = \frac{1}{\varphi(H_0)} \prod_{\substack{3 \leq \ell \leq Y \\ \ell \nmid H_0}} \frac{\ell - 2}{\ell - 1}. \tag{7.1}$$

Every base in A is a nonresidue modulo every prime in this union, and no odd prime at most Y divides its predecessor.

Lemma 7.2. *For every positive integer T, if $Y = 2T^2 + 1$, then*

$$\lambda_{A,Y} \geq \frac{1}{2\varphi(H_0)T}.$$

Proof. Adding the missing factors indexed by odd integers decreases the product, hence

$$\varphi(H_0)\lambda_{A,Y} \geq \prod_{j=1}^{T^2} \frac{2j-1}{2j} = \frac{\binom{2T^2}{T^2}}{4^{T^2}} \geq \frac{1}{2T}.$$

For the last inequality, $a_m = \binom{2m}{m}/4^m$ has $a_1 = 1/2$ and $a_{m+1}/a_m = (2m+1)/(2m+2) \geq \sqrt{m/(m+1)}$. Induction gives $a_m \geq 1/(2\sqrt{m})$. □

Proof of Theorem 1.1. Fix $\eta = \delta/4$, $\alpha = 1 - \eta$, $\beta = 1 - \eta/2$, and $\theta = \log(\beta/\alpha) > 0$. Fix Φ, m_Φ as in Corollary 4.2. Apply that corollary to each of the finitely many classes in (7.1). The total candidate weight is

$$(m_\Phi \theta \lambda_{A,Y} + o_{A,Y}(1)) \frac{x}{\log x}.$$

For large x these primes lie in $(x, 2x]$ and satisfy

$$p - 1 = Qr, \qquad x^{1-\eta} < Q \leq x^{1-\eta/2}, \qquad r < 2x^\eta.$$

Proposition 6.3 removes every possible order obstruction. Since $0 \leq \Phi \leq 1$, the weighted loss is at most the unweighted number of exceptions. Therefore

$$N_A(x) \geq \left(m_\Phi \theta \lambda_{A,Y} - \frac{C_A}{Y} + o_{A,Y}(1) \right) \frac{x}{\log x}. \tag{7.2}$$

Choose an integer $T \geq 1$ such that

$$T \geq \frac{4C_A\varphi(H_0)}{m_\Phi \theta}, \qquad Y = 2T^2 + 1 > \max_{a \in A} q_a.$$

Lemma 7.2 gives

$$m_\Phi \theta \lambda_{A,Y} \geq \frac{m_\Phi \theta}{2\varphi(H_0)T}, \qquad \frac{C_A}{Y} < \frac{C_A}{2T^2} \leq \frac{m_\Phi \theta}{8\varphi(H_0)T}.$$

Keep T, Y fixed. For sufficiently large x, the error in (7.2) has absolute value at most $m_\Phi\theta/(8\varphi(H_0)T)$. Hence (1.2) holds with

$$c_A = \frac{m_\Phi \theta}{4\varphi(H_0)T} > 0.$$

All arithmetic parameters are fixed before $x \to \infty$. The auxiliary limits inside the fixed-class statistic are taken in the order $x \to \infty$, then the number of arrays, then h.

Conversely, if an odd-cardinality subset $B \subseteq A$ has square product, and all its elements were primitive roots modulo an odd prime p, Euler's criterion would give

$$1 = \left(\frac{\prod_{a\in B} a}{p}\right) = \prod_{a\in B}\left(\frac{a}{p}\right) = (-1)^{|B|} = -1,$$

a contradiction. A prime dividing one of the bases cannot make that base a primitive root. □

Corollary 7.3. *For every A satisfying (1.1), under the same inputs,*

$$\liminf_{X\to\infty} \frac{\#\{p \le X : p \nmid \prod_{a\in A} a,\ \operatorname{ord}_p(a) = p-1 \text{ for all } a \in A\}}{\pi(X)} > 0.$$

Proof. The numerator is at least $N_A(X/2)$ for large X. Use (1.2) and $\pi(X) \sim X/\log X$. □

For $A = \{2, 3\}$ one may use $p \equiv 5 \pmod{24}$, and for $A = \{3, 6\}$ one may use $p \equiv 17 \pmod{24}$. The respective filters on $n = p - 1$ are

$$(\mathbf{1}_{4|n} - \mathbf{1}_{8|n}) \prod_{3\le \ell \le Y} (1 - \mathbf{1}_{\ell|n}), \qquad \mathbf{1}_{8|n} \prod_{3\le \ell \le Y} (1 - \mathbf{1}_{\ell|n}).$$

Both have density $\frac{1}{4}\prod_{3\le\ell\le Y}(\ell-2)/(\ell-1)$. The set $\{2, 3, 6\}$ fails (1.1), since $2 \cdot 3 \cdot 6 = 36$.